

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

Scope: fossil fuels & industry

1750–2024

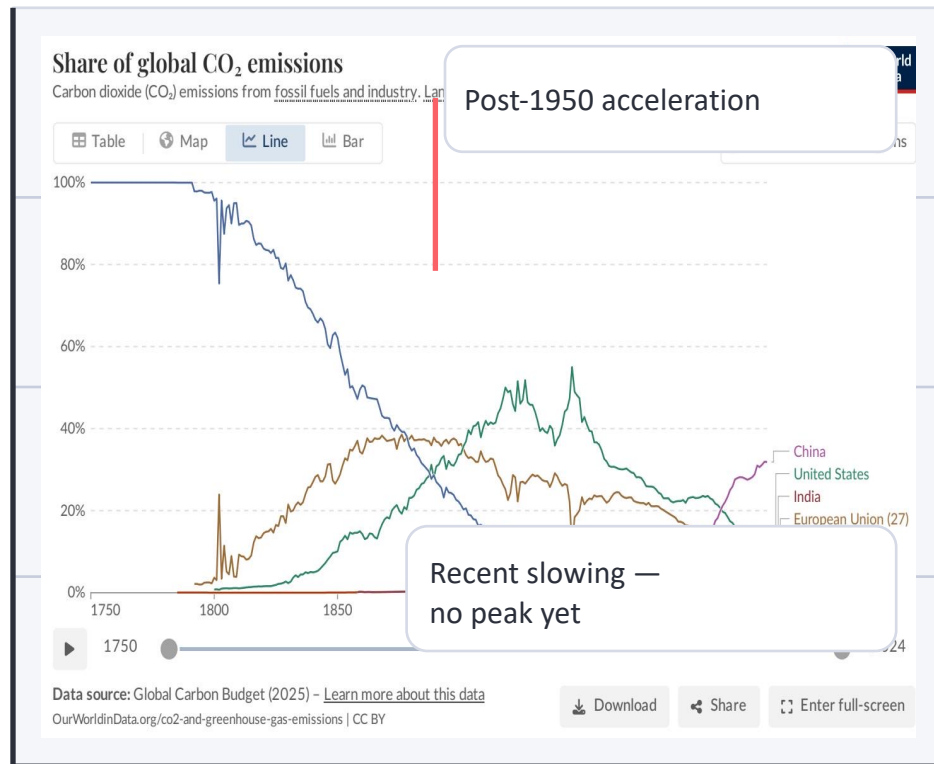
For policy analysts, climate negotiators, and sustainability researchers

Data: Global Carbon Budget (2025) via Our World in Data

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

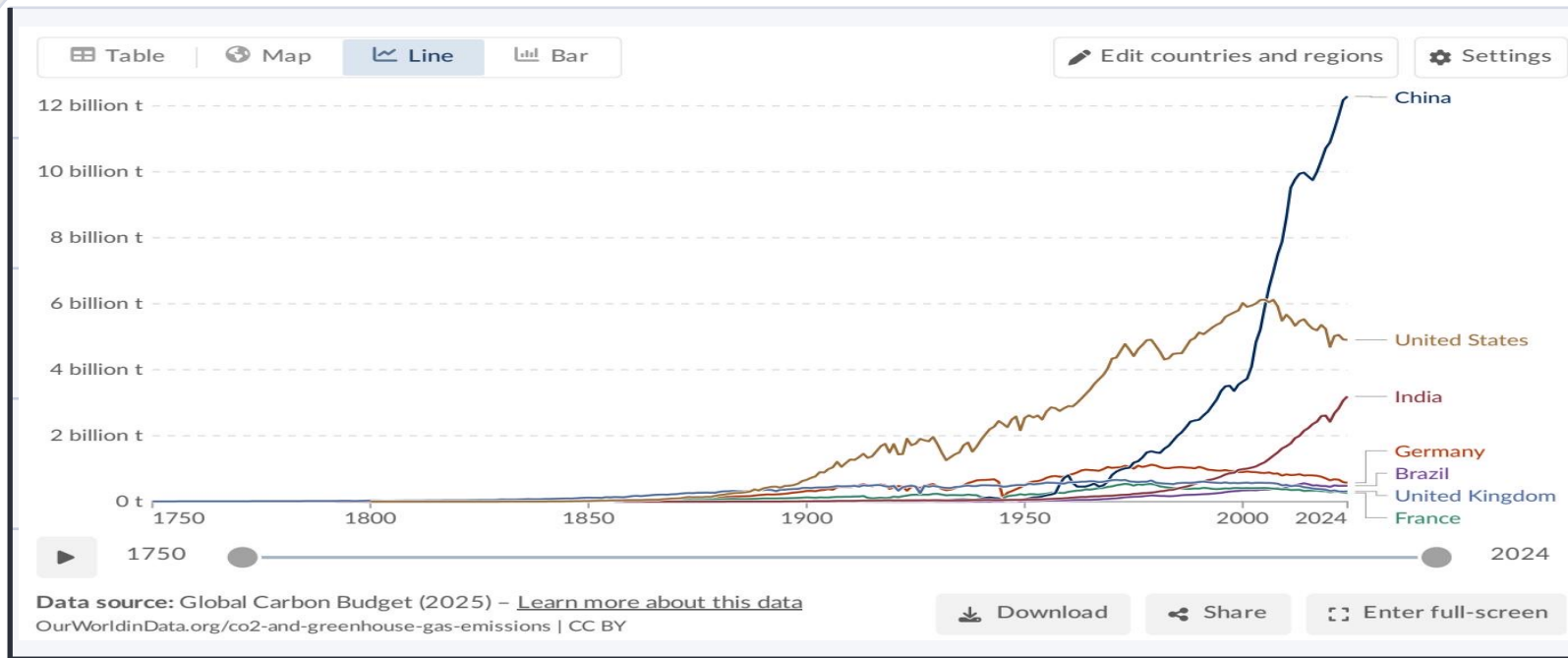
What the long-run curve shows

- Pre-industrial: negligible emissions
- 1950: ~6 Bt CO₂/yr
- 1990: >20 Bt CO₂/yr
- Today: >35 Bt CO₂/yr
- Growth slowed recently — but no peak yet



Note: Land-use emissions declined slightly; fossil fuels dominate totals.

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



Shift in shares reshapes responsibility narratives: historical dominance (Europe/US) vs. today's annual emissions (Asia).

China Now Emits More Than One-Quarter of Global CO₂

China

~31%

Rising, growth slowing

United States

~13%

Declining

India

~8%

Rising

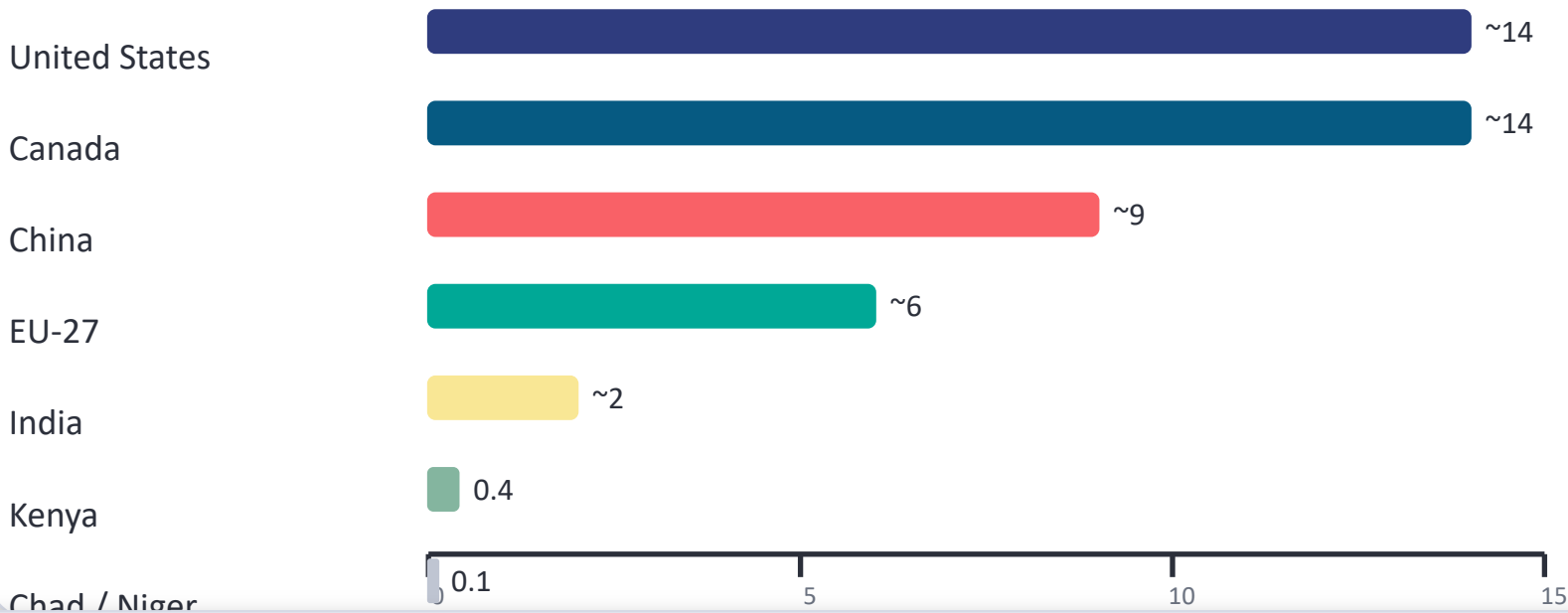
EU-27

~6%

Other notable shares (2024): Russia ~5%; Africa total ~3–4%; South America total ~3–4%; Int'l aviation & shipping ~3%.

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO₂ emissions (approx. 2024, t/person)

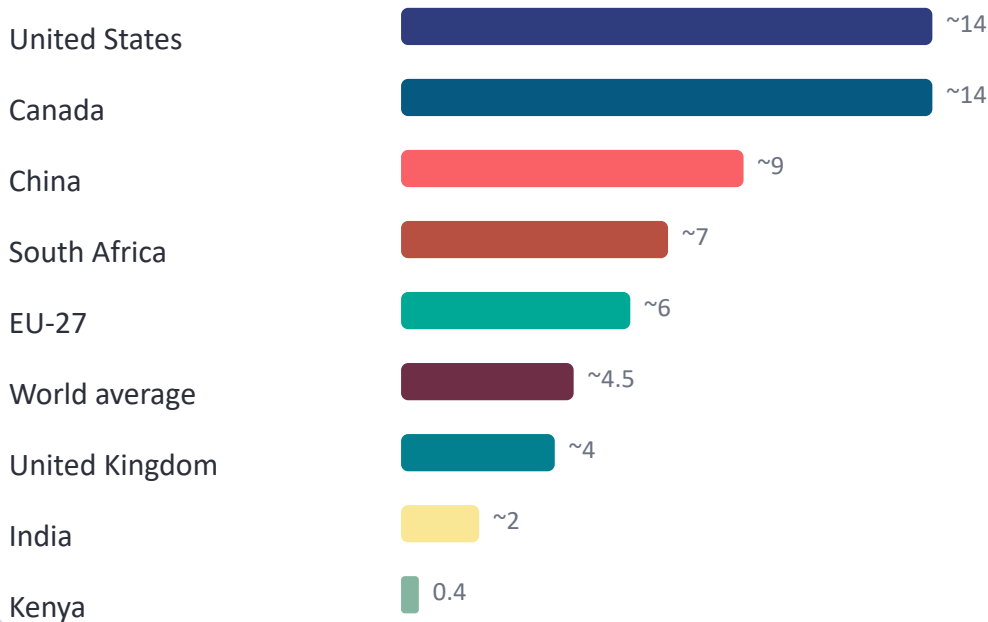


Scale of inequality: ~0.1 t/person vs ~14–15 t/person (≈150× gap).

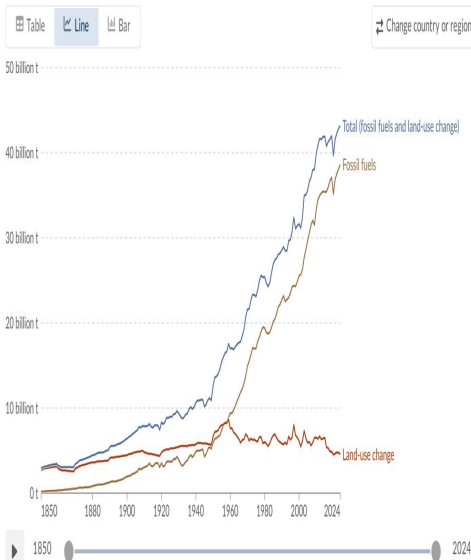
Even among wealthy countries, per capita emissions vary substantially (energy mix matters).

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Per capita CO₂ emissions (approx. 2024, t/person)

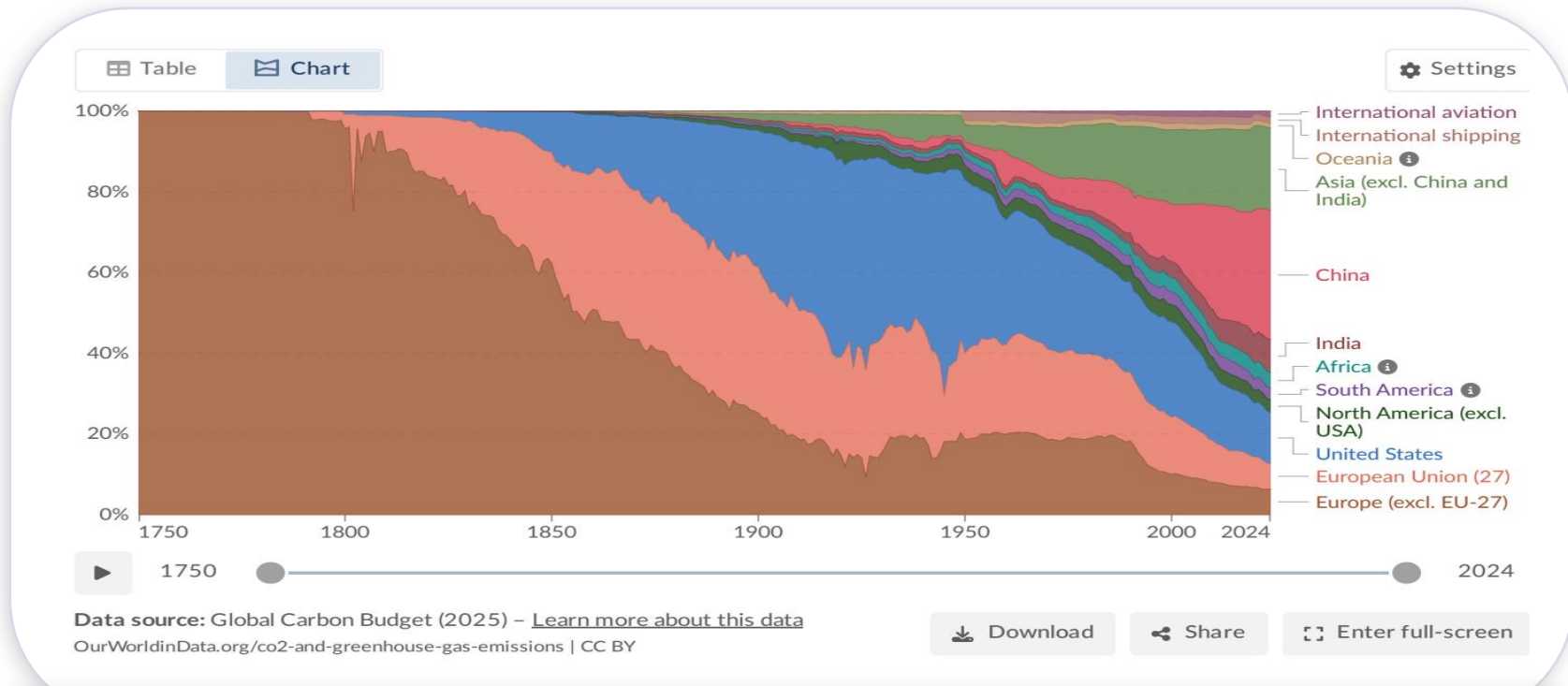


Context over time



China remains below the US per capita despite being the largest total emitter.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Headline: declines across many Western economies; increases in parts of Asia, the Middle East, and Africa offset some gains.

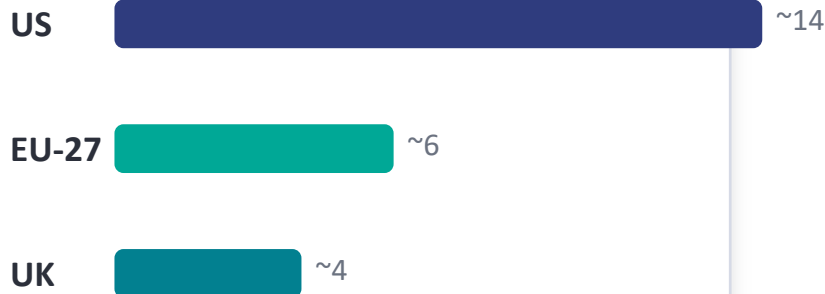
Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Among similarly wealthy countries, policy + energy systems drive large differences.

- Lower per-capita emitters: France, UK, Portugal
- Higher per-capita emitters: Germany, Netherlands, US
- Key driver: larger share of electricity from nuclear + renewables vs fossil fuels

This slide uses available per-capita figures; it avoids a detailed energy breakdown.

Per-capita comparison (approx. 2024, t/person)



Qualitative comparison (no numeric values in brief):
France/Portugal/UK < Germany/Netherlands/US
— driven by electricity mix (nuclear + renewables vs fossil fuels).

Key Takeaways: Responsibility Is Shared but Unequal

- Global emissions exceed 35 Bt/yr and have not yet peaked (fossil fuels & industry).
- China is the largest annual emitter (~31%), but per capita (~9 t) remains below the US (~14 t).
- Historical responsibility was dominated by Europe and the US (>90% in 1900), but their share is now <1/3.
- Per-capita gaps of ~150× persist between lowest emitters (~0.1 t) and high-income, high-emitting countries (~14–15 t).
- Energy policy and technology choices can decouple prosperity from emissions (e.g., UK vs US per capita).